Saptarshi Das

Quantitative Analyst

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Professional Summary

Quantitative Analyst with 2 years of experience in systematic trading, derivatives, and financial data science. Designs and backtests momentum, mean-reversion, and options strategies for NSE and MCX markets, builds event-driven backtesting and live execution systems in Python, and prices derivatives using Black-Scholes and binomial models. Experienced in deploying data pipelines on GCP and in AI-assisted development with **Claude**.

Technical Skills

Programming:	Python (NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Plotly), C++, Mojo, R, SQL, Bash, VBA
Trading:	Algorithmic Trading, Event-Driven Backtesting, Momentum and Mean-Reversion, Volatility Modelling
Derivatives:	Options Pricing (Black-Scholes, Binomial), Implied Volatility, Greeks, Delta Hedging, Risk Management
Data Science:	Machine Learning, Time Series Forecasting, Regression and Classification, Feature Engineering, Statistics
Platforms:	GCP (BigQuery, Cloud Storage, Cloud Scheduler), ETL Pipelines, Broker APIs, Market Data Feeds
AI and Tools:	**Claude** (AI-Assisted Development), Streamlit, Power BI, Tableau, Excel (VBA Macros), Git, Linux

Professional Experience

Quantitative Analyst | Jun 2025 – Present — **Cloudcraftz AI Pvt. Ltd.** | Kolkata, India

Systematic Trading and Backtesting Engine — *Python, Machine Learning, REST APIs*

- Architected a modular, event-driven backtesting engine with a pluggable adapter layer, enabling a single codebase to run backtest, paper, and live trading across NSE equities and MCX commodities with realistic transaction-cost modelling.
- Backtested momentum and mean-reversion strategies (EMA crossovers, RSI filters, VWAP deviation) on minute-level data spanning **49 instruments and 106 trading days**; automated live order execution through broker REST APIs with retry handling and structured logging.

Options Strategy Research and Risk Management — *Python, Options Greeks, Volatility Models*

- Researched and backtested a **VWAP mean-reversion and Bollinger-squeeze options strategy** on MCX crude oil, combining volatility and trend filters with **delta-targeted strike selection** to deliver a **30% ROI**.
- Designed a volatility-regime **ATM straddle strategy** using an ATR fast/slow ratio and intraday implied-volatility z-score, with regime-based position sizing, spot-move circuit breakers, and daily loss limits.
- Developed an **adaptive hedging framework** that monitors position Greeks (**delta, gamma, vega**) in real time and triggers rebalancing on regime shifts or risk-limit breaches, reducing unhedged exposure in volatile markets.

Options Analytics Pipeline and Trade Operations — *Python, GCP, BigQuery*

- Built **OptiStream**, an end-to-end options data pipeline computing implied volatility and the **full Greeks surface (delta, gamma, theta, vega, rho)** via Black-Scholes and binomial models; deployed on GCP with BigQuery and Cloud Scheduler, reducing manual processing effort by **90%**.
- Automated **trade reconciliation**, flagging position breaks, slippage anomalies, and execution shortfalls in real time and reducing manual reconciliation effort by **80%**.

Real-Time Trading Dashboard and Live Deployment — *Mojo, Python, Streamlit, Claude*

- Developed a real-time monitoring dashboard with a compiled **Mojo** analytics backend for trade-log parsing and **P&L aggregation**, displaying intraday P&L, open positions, and signal history on a 30-second auto-refresh; built end-to-end using **Claude for AI-assisted development**.
- Deployed the trading engine for automated live and paper execution on a scheduled daily cycle, integrating broker market-data feeds, tick recording, session logging, and email trade alerts.

Management Trainee, Finance and Analytics | Dec 2023 – Jul 2024 — **K12 Techno Services** | Bengaluru, India

- Conducted financial due diligence on multi-crore transactions using DCF and relative valuation (EV/EBITDA, P/E); developed a Python **logistic regression model** predicting customer default rates to enable credit-risk segmentation.
- Automated financial reporting pipelines using **Power BI and SQL** and developed **time-series forecasting models** to support investment decisions with quantified risk parameters.

Education

MBA, Finance and Analytics | Jul 2021 – May 2023 — **IIM Amritsar** | GPA: 76.5%

Coursework: Financial Modelling, Derivatives Pricing, Portfolio Management, Risk Management

B.Tech, Mechanical Engineering | Jul 2015 – May 2019 — **Meghnad Saha Institute of Technology** | GPA: 74.5%

Certifications and Achievements

Algorithmic Trading with Python, Machine Learning and AWS, Udemy (2025) | Portfolio and Risk Management, ISB | Kaggle Certifications: Python, Machine Learning, NLP | 3rd Place, InSight Out Valuation Competition, IIM Amritsar (2023)